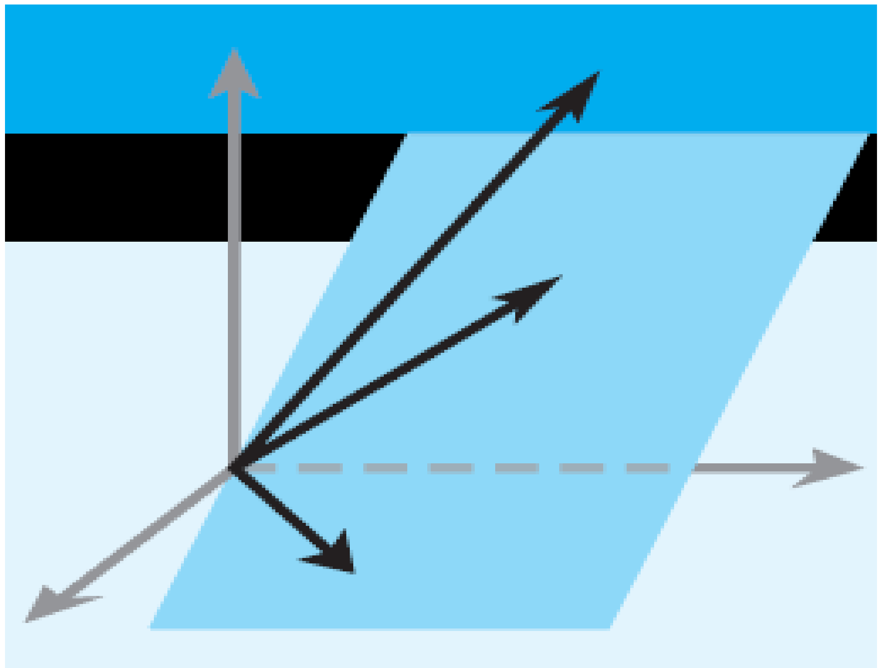
Elementary Linear Algebra



Chapter 6

Howard Anton

Chapter 6 Inner Product Spaces

- 6.1 Inner Products
- 6.2 Angle and Orthogonality in Inner Product Spaces

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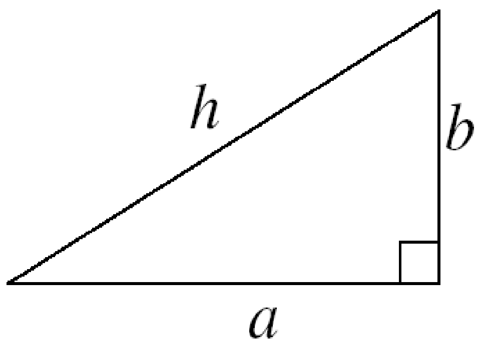
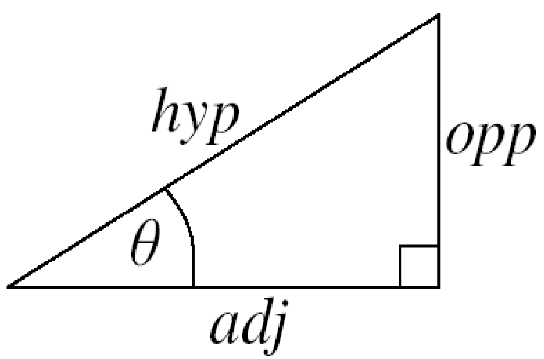
- 6.3 Gram-Schmidt Process; QR-Decomposition
- 6.4 Best Approximation; Least Squares
- 6.5 Least Squares Fitting to Data
- 6.6 Function Approximation; Fourier Series

Section 6.2
Angle and Orthogonality in Inner Product Spaces

Review





<u>Pythagoras's Theorem</u>

$$a^2 + b^2 = h^2$$

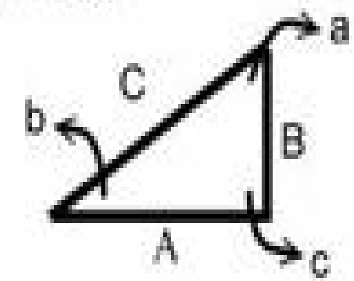
<u>Trigonometric Ratios</u>

$$\sin(\theta) = \frac{opp}{hyp}$$

$$\cos(\theta) = \frac{adj}{hyp}$$

$$\tan(\theta) = \frac{opp}{adj}$$

直角三角形



A,B,C 為三角形三個邊
a,b,c 為三角形三個角
其中 c 為直角(90度)；所以 $a+b=90$

畢氏定理 $C^2 = A^2 + B^2$

正弦定理 $\left(\frac{\text{對邊}}{\text{斜邊}}\right)$ $\sin a = \frac{A}{C}$, $\sin b = \frac{B}{C}$

餘弦定理 $\left(\frac{\text{鄰邊}}{\text{斜邊}}\right)$ $\cos a = \frac{B}{C}$, $\cos b = \frac{A}{C}$

正切定理 $\left(\frac{\text{對邊}}{\text{鄰邊}}\right)$ $\tan a = \frac{A}{B}$, $\tan b = \frac{B}{A}$

Section 6.2
Angle and Orthogonality in Inner Product Spaces

Review

R^2

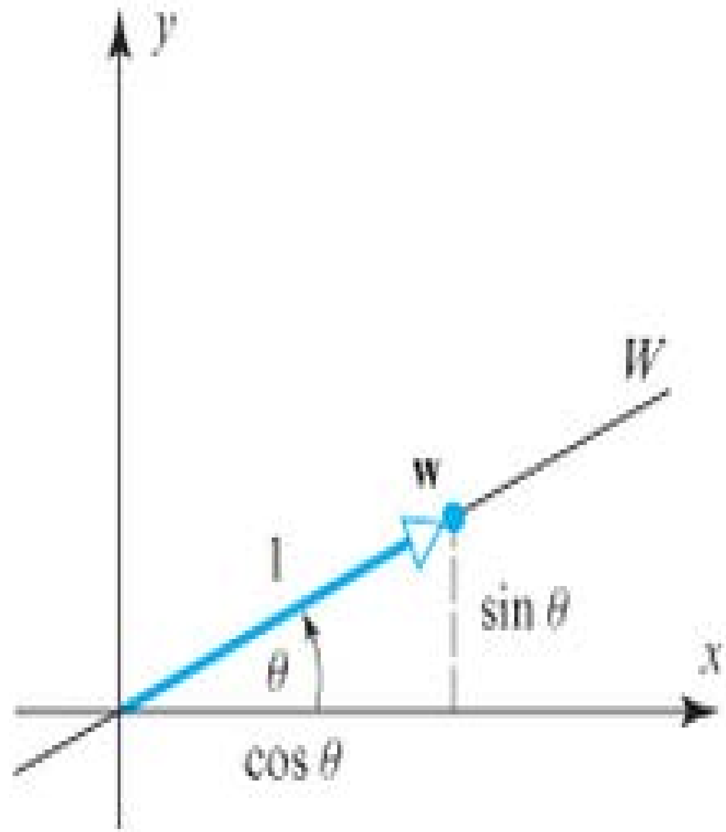


Figure 6.4.2

Section 6.2
Angle and Orthogonality in Inner Product Spaces

Review Law of cosine

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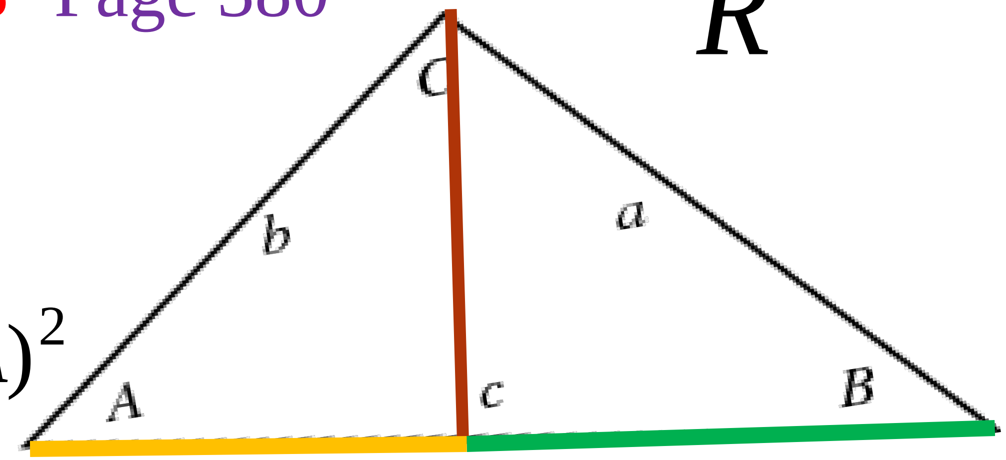
R^2



length of red line = $b^2 - b^2 \cos^2 A$

length of red line = $a^2 - (c - b\cos A)^2$

$b\cos A$ $\quad$ $c - b\cos A$

$\cos A = adj / b$

$adj = b\cos A$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$b^2 - b^2\cos^2 A = a^2 - c^2 + 2bc\cos A - b^2\cos^2 A$$

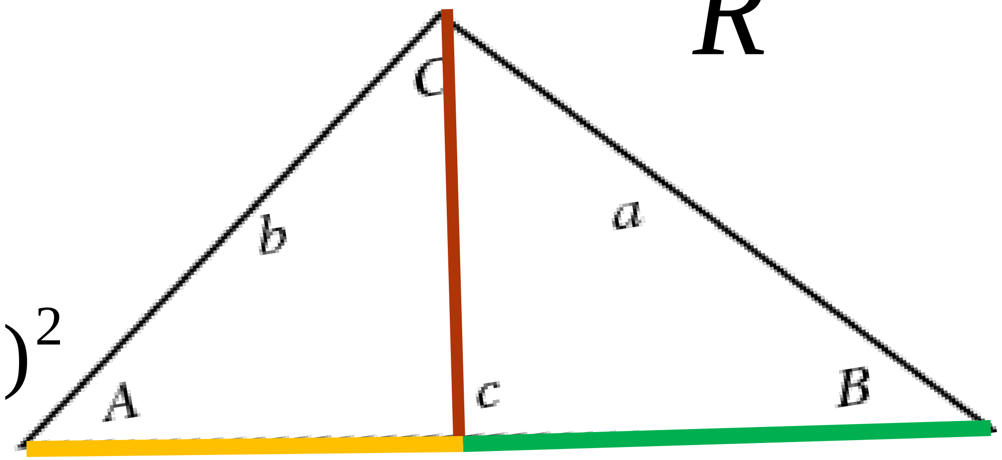
Section 6.2
Angle and Orthogonality in Inner Product Spaces

Review Law of cosine

R^2

length of red line = $a^2 - a^2\cos^2 B$

length of red line = $b^2 - (c - a\cos B)^2$



$c - a\cos B$ $\quad$ $a\cos B$

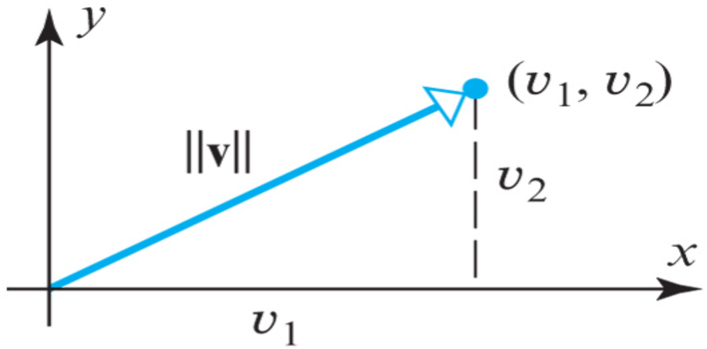
$$\cos B = adj / a$$
$$adj = a\cos B$$

$$b^2 = a^2 + c^2 - 2ac\cos B$$

$$a^2 - a^2\cos^2 B = \quad b^2 - c^2 + 2ac\cos B - a^2\cos^2 B$$

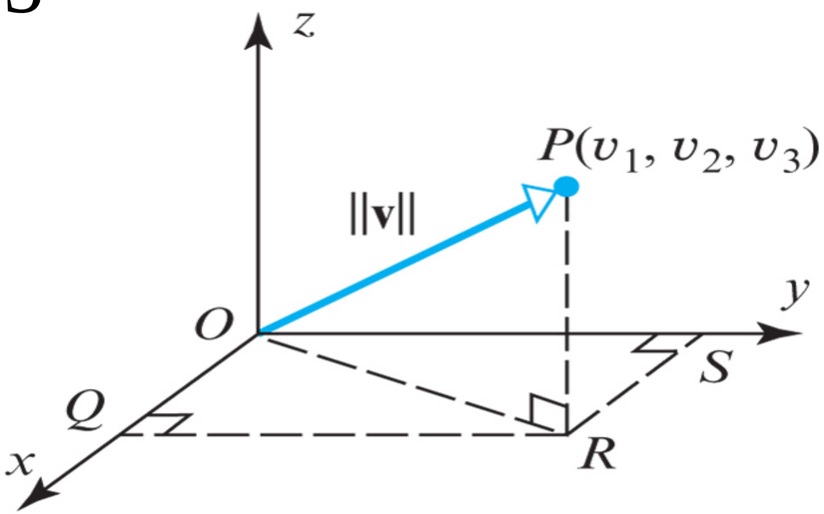
Section 6.2
Angle and Orthogonality in Inner Product Spaces

R^2



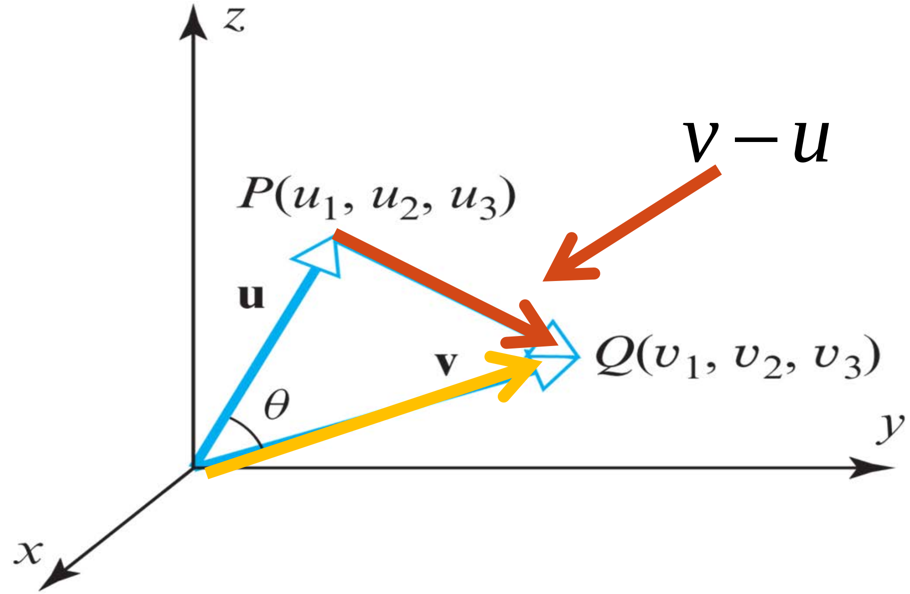
(a)

R^3



(b)

R^3



$$a^2 = b^2 + c^2 - 2bc\cos A$$

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$$\|v-u\|^2 = \|u\|^2 + \|v\|^2 - 2\|v\|\|u\|\cos\theta$$

Section 6.2
Angle and Orthogonality in Inner Product Spaces

$$R^3$$

$$\|v-u\|^2 = \|v\|^2 + \|u\|^2 - 2\|v\|\|u\|\cos\theta$$

$$u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \quad v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$\|u\| = \sqrt{u_1^2 + u_2^2 + u_3^2} \quad \|u\|^2 = u_1^2 + u_2^2 + u_3^2$$

$$\|v\| = \sqrt{v_1^2 + v_2^2 + v_3^2} \quad \|v\|^2 = v_1^2 + v_2^2 + v_3^2$$

$$\|v-u\| = \sqrt{(v_1-u_1)^2 + (v_2-u_2)^2 + (v_3-u_3)^2}$$

Key point

Section 6.2
Angle and Orthogonality in Inner Product Spaces

$$\|v-u\|^2 - \|v\|^2 - \|v\|^2 = -(2v_1u_1 + 2v_2u_2 + 2v_3u_3) \qquad R^3$$

$$-2\|u\|\|v\|\cos\theta \quad = -(2v_1u_1 + 2v_2u_2 + 2v_3u_3)$$

$$\|u\|\|v\|\cos\theta \quad = \langle u, v\rangle = u \bullet v$$

$$\cos\theta = \frac{<u, v>}{\|u\|\|v\|} \Rightarrow \theta = \cos^{-1}\left(\frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|\,\|\mathbf{v}\|}\right)$$

Section 6.2
Angle and Orthogonality in Inner Product Spaces

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DEFINITION 3

If $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors in R^2 or R^3, and if θ is the angle between $\mathbf{u}$ and $\mathbf{v}$, then the ***dot product*** (also called the ***Euclidean inner product***) of $\mathbf{u}$ and $\mathbf{v}$ is denoted by $\mathbf{u} \cdot \mathbf{v}$ and is defined as

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta \tag{12}$$

$$\cos\theta = \frac{< u, v >}{\|u\| \|v\|}$$

If $\mathbf{u} = \mathbf{0}$ or $\mathbf{v} = \mathbf{0}$, then we define $\mathbf{u} \cdot \mathbf{v}$ to be 0.

Section 6.2
Angle and Orthogonality in Inner Product Spaces

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DEFINITION 4

If $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ are vectors in R^n, then the ***dot product*** (also called the ***Euclidean inner product***) of $\mathbf{u}$ and $\mathbf{v}$ is denoted by $\mathbf{u} \cdot \mathbf{v}$ and is defined by

$$\boxed{\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n} \tag{17}$$

Section 6.2
Angle and Orthogonality in Inner Product Spaces

EXAMPLE 1 **Cosine of an Angle Between Two Vectors in R^4**

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Let R^4 have the Euclidean inner product. Find the cosine of the angle θ between the vectors $\mathbf{u} = (4, 3, 1, -2)$ and $\mathbf{v} = (-2, 1, 2, 3)$.

Solution We leave it for you to verify that

$$\|u\| = \sqrt{30}, \quad \|v\| = \sqrt{18}, \quad \langle u, v\rangle = -9$$

from which it follows that

$$\cos\theta = \frac{\langle u, v\rangle}{\|u\| \; \|v\|} = -\frac{9}{\sqrt{30}\sqrt{18}} = -\frac{3}{2\sqrt{15}}$$

Section 6.2
Angle and Orthogonality in Inner Product Spaces

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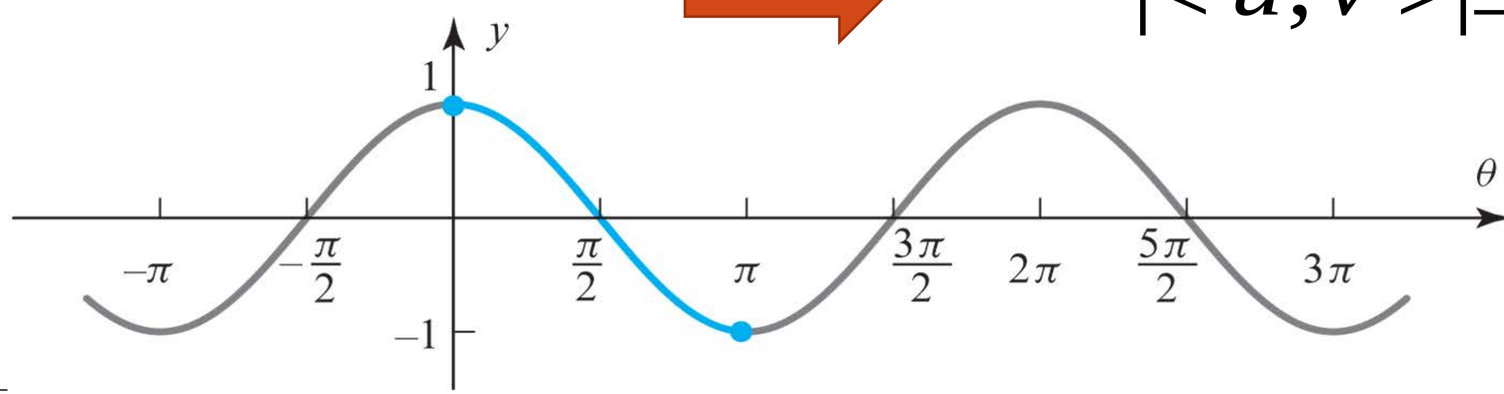
R^3

$$\theta = \cos^{-1}\left(\frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|\,\|\mathbf{v}\|}\right)$$

Key point

$$\cos\theta = \frac{<u,v>}{\|u\|\|v\|} \Rightarrow -1 \le \frac{<u,v>}{\|u\|\|v\|} \le 1$$

$$\Rightarrow |<u,v>| \le \|u\|\|v\|$$



Section 6.2
Angle and Orthogonality in Inner Product Spaces

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THEOREM 6.2.1 Cauchy-Schwarz Inequality

If $\mathbf{u}$ *and* $\mathbf{v}$ *are vectors in a real inner product space* V, *then*

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \le \|\mathbf{u}\| \, \|\mathbf{v}\| \tag{3}$$

For general vector space,

the proof is based on a clever trick

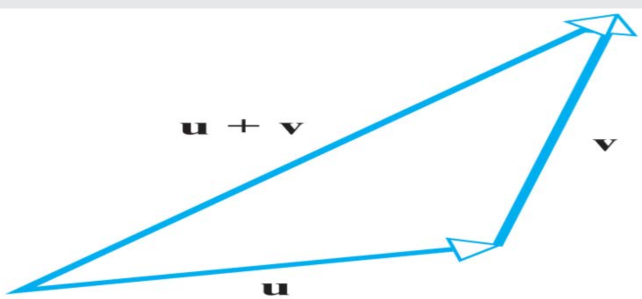
Section 6.2
Angle and Orthogonality in Inner Product Spaces

THEOREM 6.2.2

If $\mathbf{u}, \mathbf{v}$, *and* $\mathbf{w}$ *are vectors in a real inner product space* V, *and if* k *is any scalar, then:*

(a) $\|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$ [Triangle inequality for vectors]

(b) $d(\mathbf{u}, \mathbf{v}) \le d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$ [Triangle inequality for distances]



$\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$

Key point

Section 6.2

Angle and Orthogonality in Inner Product Spaces

Show that $$\|u+v\| \le \|u\| + \|v\|$$

Proof:

$$\|u+v\|^2 = \langle u+v, u+v \rangle$$ By definition

$$= \langle u,u \rangle + \langle u,v \rangle + \langle v,u \rangle + \langle v,v \rangle$$

$$= \langle u,u \rangle + \langle u,v \rangle + \langle u,v \rangle + \langle v,v \rangle$$

$$= \|u\|^2 + 2\langle u,v \rangle + \|v\|^2$$

$$\le \|u\|^2 + 2|\langle u,v \rangle| + \|v\|^2$$ Property of absolute value

$$\le \|u\|^2 + 2\|u\|\|v\| + \|v\|^2$$ Cauchy Schwarz inequality

$$= (\|u\| + \|v\|)^2$$

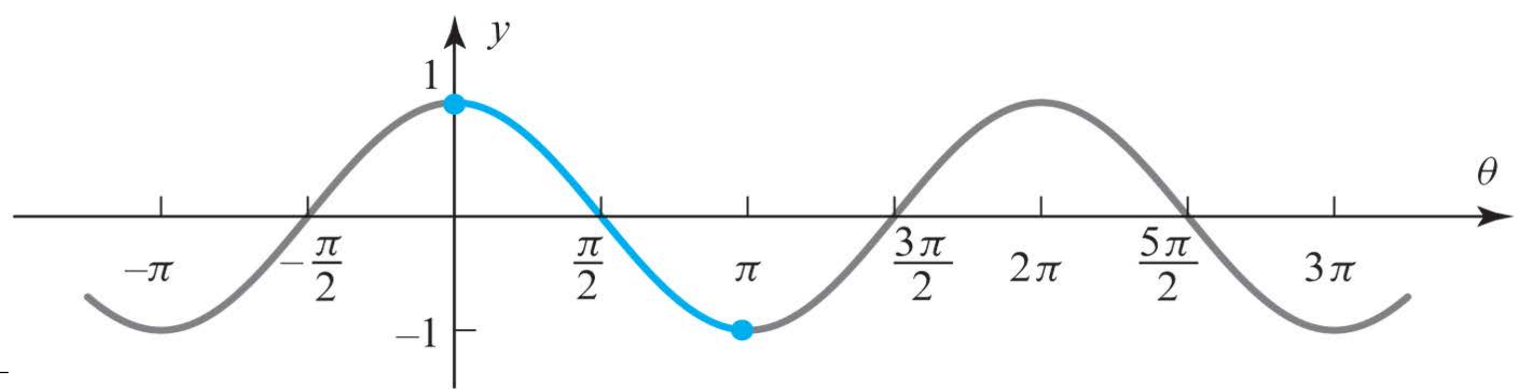
Section 6.2
Angle and Orthogonality in Inner Product Spaces

DEFINITION 1 Two vectors **u** and **v** in an inner product space are called ***orthogonal*** if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Key point

$$R^3$$

$$\langle u, v \rangle = u \bullet v = \|u\|\|v\| \cos\theta \quad = 0$$



Section 6.2
Angle and Orthogonality in Inner Product Spaces

EXAMPLE 2 Orthogonality Depends on the Inner Product

The vectors $\mathbf{u} = (1, 1)$ and $\mathbf{v} = (1, -1)$ are orthogonal with respect to the Euclidean inner product on R^2 since

$$\mathbf{u} \cdot \mathbf{v} = (1)(1) + (1)(-1) = 0$$

However, they are not orthogonal with respect to the weighted Euclidean inner product $\langle \mathbf{u}, \mathbf{v} \rangle = 3u_1v_1 + 2u_2v_2$ since

$$\langle \mathbf{u}, \mathbf{v} \rangle = 3(1)(1) + 2(1)(-1) = 1 \neq 0$$

Section 6.2
Angle and Orthogonality in Inner Product Spaces

EXAMPLE 3 Orthogonal Vectors in M_{22}

If M_{22} has the inner product of Example 6 in the preceding section, then the matrices

$$U = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad V = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$$

$$\langle U, V \rangle = trace(U^T V)$$

are orthogonal since

$$\langle U, V \rangle = 1(0) + 0(2) + 1(0) + 1(0) = 0$$

Section 6.2
Angle and Orthogonality in Inner Product Spaces

EXAMPLE 4 Orthogonal Vectors in P_2

Let P_2 have the inner product

$$< \mathrm{p}, \mathrm{q} >= \int_{-1}^{1} p(x)q(x)dx$$

$\mathrm{p} = x, \mathrm{q} = x^2$ Then

$$\|\mathbf{p}\| = \langle \mathbf{p}, \mathbf{p} \rangle^{1/2} = \left[\int_{-1}^{1} xx\,\mathrm{dx}\right]^{1/2} = \left[\int_{-1}^{1} x^2 dx\right]^{1/2} = \sqrt{\frac{2}{3}}$$

$$\|\mathbf{q}\| = \langle \mathbf{q}, \mathbf{q} \rangle^{1/2} = \left[\int_{-1}^{1} x^2x^2 dx\right]^{1/2} = \left[\int_{-1}^{1} x^4 dx\right]^{1/2} = \sqrt{\frac{2}{5}}$$

$$\langle \mathbf{p}, \mathbf{q} \rangle = \int_{-1}^{1} xx^2 dx = \int_{-1}^{1} x^3 dx = 0$$

Because $\langle \mathbf{p}, \mathbf{q} \rangle = 0$, the vectors $\mathbf{p} = x$ and $\mathbf{q} = x^2$ are orthogonal relative to the given inner product.

Section 6.2
Angle and Orthogonality in Inner Product Spaces

THEOREM 6.2.3 Generalized Theorem of Pythagoras

If $\mathbf{u}$ *and* $\mathbf{v}$ *are orthogonal vectors in a real inner product space, then*

$$\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

Proof:

$$\|u+v\|^2 = <\mathrm{u}+\mathrm{v}, \mathrm{u}+\mathrm{v}>$$

By definition

$$= <\mathrm{u},\mathrm{u}> + <\mathrm{u},\mathrm{v}> + <\mathrm{v},\mathrm{u}> + <\mathrm{v},\mathrm{v}>$$

$$= <\mathrm{u},\mathrm{u}> + <\mathrm{u},\mathrm{v}> + <\mathrm{u},\mathrm{v}> + <\mathrm{v},\mathrm{v}>$$

$$= \|\mathrm{u}\|^2 + \underbrace{2<\mathrm{u},\mathrm{v}>}_{0} + \|\mathrm{v}\|^2$$

Section 6.2
Angle and Orthogonality in Inner Product Spaces

EXAMPLE 5 Theorem of Pythagoras in P_2

In Example 4 we showed that $\mathbf{p} = x$ and $\mathbf{q} = x^2$ are orthogonal with respect to the inner product

$$\langle \mathbf{p}, \mathbf{q} \rangle = \int_{-1}^{1} p(x)\, q(x)\, dx$$

$$< \mathrm{p}, \mathrm{q} >= \int_{-1}^{1} p(x)q(x)dx$$

Section 6.2
Angle and Orthogonality in Inner Product Spaces

on P_2. It follows from Theorem 6.2.3 that

$$\|\mathbf{p}+\mathbf{q}\|^2=\|\mathbf{p}\|^2+\|\mathbf{q}\|^2$$

Thus, from the computations in Example 4, we have

$$\|\mathbf{p}+\mathbf{q}\|^2=\left(\sqrt{\frac{2}{3}}\right)^2+\left(\sqrt{\frac{2}{5}}\right)^2=\frac{2}{3}+\frac{2}{5}=\frac{16}{15}$$

We can check this result by direct integration:

$$\begin{aligned}\|\mathbf{p}+\mathbf{q}\|^2 &= \langle \mathbf{p}+\mathbf{q},\mathbf{p}+\mathbf{q}\rangle=\int_{-1}^{1}(x+x^2)\,(x+x^2)\,dx\\ &= \int_{-1}^{1}x^2dx+2\int_{-1}^{1}x^3dx+\int_{-1}^{1}x^4dx=\frac{2}{3}+0+\frac{2}{5}=\frac{16}{15}\end{aligned}$$

Section 6.2
Angle and Orthogonality in Inner Product Spaces

DEFINITION 2

If W is a subspace of a real inner product space V, then the set of all vectors in V that are orthogonal to every vector in W is called the ***orthogonal complement*** of W and is denoted by the symbol $W^{\perp}$.

Section 6.2
Angle and Orthogonality in Inner Product Spaces

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THEOREM 6.2.4

If W is a subspace of a real inner product space V, then:

(a) $W^{\perp}$ *is a subspace of* V.

(b) $W \cap W^{\perp} = \{\mathbf{0}\}$.

Proof:

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$$W^{\perp} = \{\mathrm{v} : <\mathrm{v}, \mathrm{u}> = 0, \quad \forall\, \mathrm{u} \in W\}$$

Discuss in the class!!

Section 6.2
Angle and Orthogonality in Inner Product Spaces

THEOREM 6.2.5

If W is a subspace of a real finite-dimensional inner product space V, then the orthogonal complement of $W^{\perp}$ is W; that is,

$$\left(W^{\perp}\right)^{\perp} = W$$

Proof:

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Angle and Orthogonality in Inner Product Spaces

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EXAMPLE 6 Basis for an Orthogonal Complement

Let W be the subspace of R^6 spanned by the vectors

$$\mathbf{w}_1 = (1, 3, -2, 0, 2, 0), \quad \mathbf{w}_2 = (2, 6, -5, -2, 4, -3),$$
$$\mathbf{w}_3 = (0, 0, 5, 10, 0, 15), \quad \mathbf{w}_4 = (2, 6, 0, 8, 4, 18)$$

Find a basis for the orthogonal complement of W.

Section 6.2
Angle and Orthogonality in Inner Product Spaces

Solution

The subspace W is the same as the row space of the matrix

$$A = \begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 \\ 0 & 0 & 5 & 10 & 0 & 15 \\ 2 & 6 & 0 & 8 & 4 & 18 \end{bmatrix}$$

Since the row space and null space of A are orthogonal complements, our problem reduces to finding a basis for the null space of this matrix. In Example 4 of Section 4.7 we showed that

$$\mathbf{v}_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -4 \\ 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Section 6.2
Angle and Orthogonality in Inner Product Spaces

form a basis for this null space. Expressing these vectors in comma-delimited form (to match that of $\mathbf{w}_1$, $\mathbf{w}_2$, $\mathbf{w}_3$, and $\mathbf{w}_4$), we obtain the basis vectors

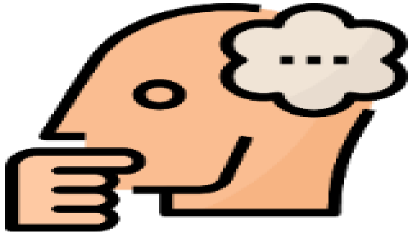
$$\mathbf{v}_1 = (-3, 1, 0, 0, 0, 0), \quad \mathbf{v}_2 = (-4, 0, -2, 1, 0, 0), \quad \mathbf{v}_3 = (-2, 0, 0, 0, 1, 0)$$

You may want to check that these vectors are orthogonal to $\mathbf{w}_1$, $\mathbf{w}_2$, $\mathbf{w}_3$, and $\mathbf{w}_4$ by computing the necessary dot products.

Homework

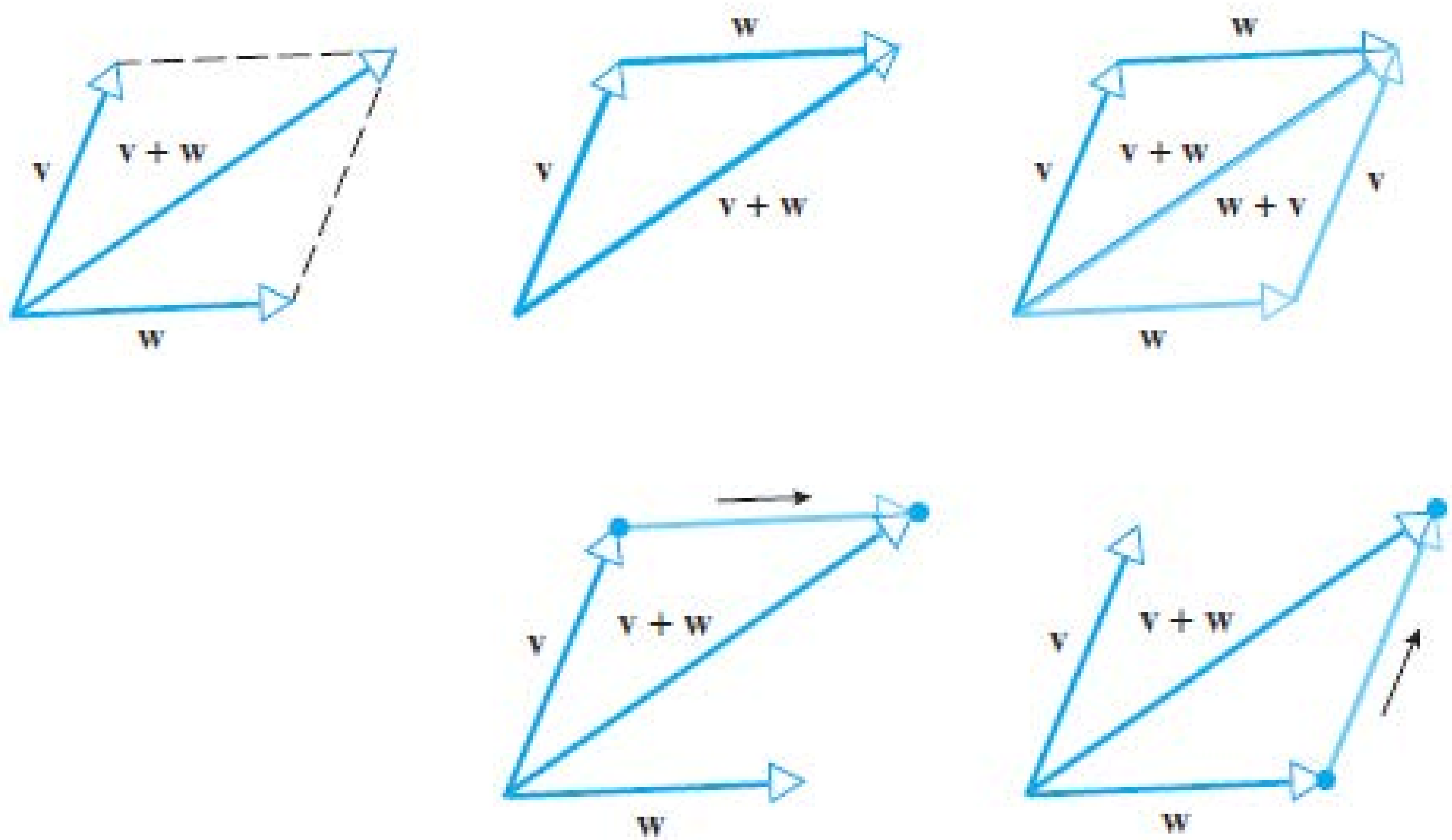
Sec 6.2

# True/False	Page 347
# 45, 48 (a)	Page 380
# 55, 56, 57	Page 380
# 61, 63, 64 (a)	Page 381



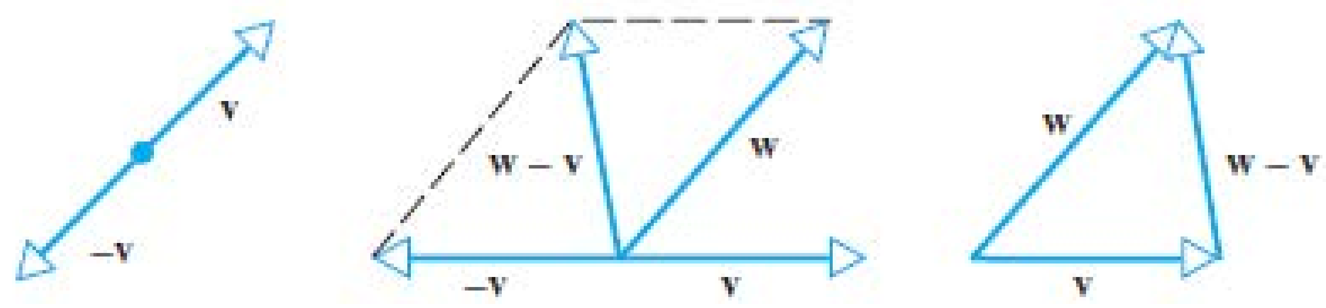
Section 3.1 Vectors **Review**

Addition of vectors by the parallelogram or triangle rules

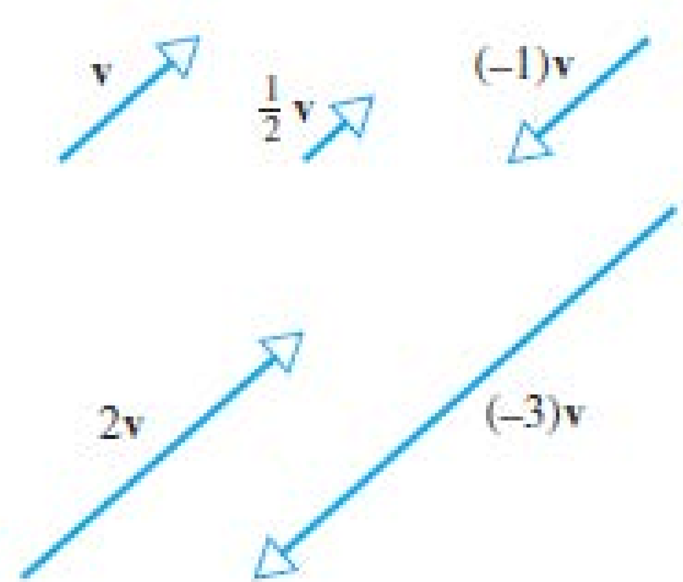


Section 3.1 Vectors **Review**

Subtraction:

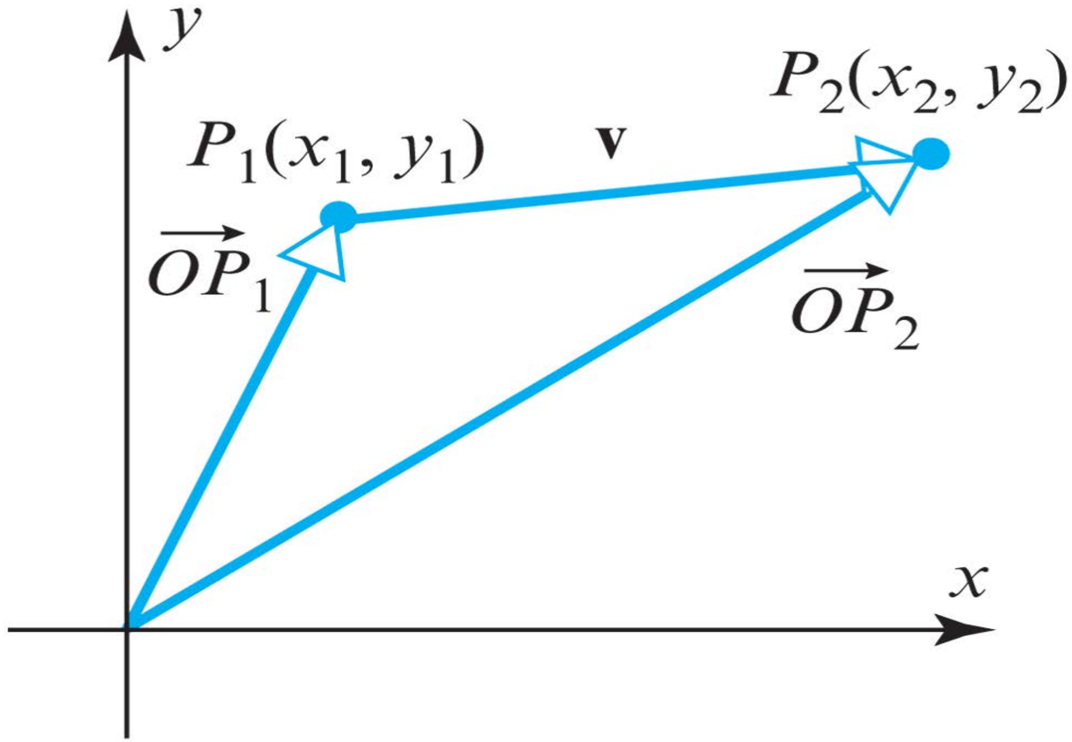


Scalar Multiplication:



Section 3.1 Vectors **Review**

Subtraction:



$$\boxed{\mathbf{v} = \overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1}}$$